

Homework Assignment 6

1. A rocket vehicle has the following characteristics:

- Initial mass 90000 kg
- Propellant mass 72000 kg
- Specific impulse 330 s

This rocket vertically ascends in constant gravity ($g = g_e = 9.81 \text{ m/s}^2$). Neglecting drag, answer the following questions:

- If the burnout acceleration is not to exceed $4.8g_e$, determine the maximum permissible mass flow rate. What is the associated burn time? {Ans.: 320 kg/s, 225 s}
 - Using the burn time calculated, find the launch acceleration of the vehicle. {Ans.: $0.17g_e$ }
 - What is the rocket acceleration just after burnout?
2. A rocket ($I_{sp} = 330 \text{ s}$) is to impart velocity rise of 7.8 km/s to a payload. The gravity and drag losses in ascent, $\overline{g \cos \theta}$ and $\left(\frac{\overline{D}}{M}\right)$, are known to be 9 m/s^2 and 0.7 m/s^2 , respectively. If the burn time is 200 s, what fraction of the initial rocket mass (M_0) is comprised of propellant? Redo the calculation assuming the launch to be impulsive. {Ans.: 0.95, 0.91}
3. A launcher has specific impulse of 360 s, payload ratio of 0.15 and structural coefficient 0.07.
- (a) Find the impulsive velocity rise provided by the launcher. {Ans.: 5841 m/s}
 - (b) If the payload mass is 1000 kg, find the mass of the launcher. {Ans.: 6667 kg}
 - (c) Find the structural mass of the launcher. {Ans.: 467 kg}
 - (d) What are initial and burnout masses of the rocket. {Ans.: 7667 kg, 1467 kg}